

ARYAN AGARWAL

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Education

University of Illinois Urbana-Champaign

Champaign, IL

Major: Computer Engineering (GPA: 3.89)

Expected Graduation: May 2028

Relevant Coursework: Autonomous Vehicles, Operating Systems, Robotic Systems, Real-Time Systems, Signal Processing, Data Structures & Algorithms, Parallel Programming

Experience

Molex

May 2026 – Aug 2026

Firmware Engineering Intern

Fremont, CA

- Automated Molex's firmware codebase migration from Keil µVision to SEGGER Embedded Studio.
- Developed **embedded C firmware** for a 1PPS synchronization system using hardware-timer capture, interrupt-driven timestamping, and EEPROM-stored calibration data to achieve **sub-nanosecond timing accuracy**.
- Enabled a **shared control loop** to support two vendor analog front ends through **runtime chip-ID detection**, **dynamic driver selection**, and **device-specific register configuration**.
- Redesigned Molex's firmware codebase to support multiple MCUs by separating shared logic from platform-specific logic.
- Reworked **FreeRTOS tasks** and added platform-specific **I2C** and **SPI** peripheral code for multi-MCU support.

Cisco

May 2025 – Aug 2025

ASIC Engineering Intern

San Jose, CA

- Measured **SDF-annotated** delays from inputs to flip-flop D-pins and clock pins across **thousands of signals** using a **Python-scripted SystemVerilog testbench**.
- Computed per-signal skew offsets and applied them through an RTL wrapper, achieving **100% gate-level vs RTL alignment**.
- Developed an **FSDB** waveform analysis tool to compare expected vs. observed signal transitions, reducing manual debug time by **35%** and accelerating root-cause analysis of gate-level timing failures.
- Designed an **HTML** report interface with searchable signal-timestamp tables, reducing waveform inspection time by **25%**.

Spatial Audio Research, UIUC

Jan 2026 – Present

Undergraduate Researcher – Advised by Prof. Yu-Lin Wei

Champaign, IL

- Kept sound sources world-locked during live head motion by combining orientation data from a **9-axis IMU** with a real-time **Python binaural renderer** using CIPIC **HRIRs** and **SciPy FFT-based convolution**.
- Reduced audible artifacts during head motion by replacing **nearest-neighbor HRIR selection** with **barycentric interpolation** across neighboring CIPIC measurements, smoothing filter updates as the listener's head orientation changed.
- Improved perceived motion-to-audio latency by using **STFT** analysis of recent IMU measurements to extrapolate head motion **100 ms** ahead and proactively update the corresponding **HRIR filters**.
- Built a time-aligned IMU + spatial-audio dataset to test world-locking accuracy, head-pose prediction, and motion-to-audio latency.

Projects

Vision-Guided UR3 Drawing Robot

- Developed an **OpenCV** vision pipeline using median filtering, edge detection, contour extraction, and polygonal simplification to convert arbitrary images into robot drawing waypoints.
- Transformed image contours into Cartesian trajectories and computed **UR3** joint commands using inverse kinematics for drawing.
- Integrated **ROS** motion control with MoveJ positioning and MoveL Cartesian tracing, publishing UR3 commands while monitoring position feedback throughout trajectory execution to accurately reproduce input images.

VisionVest - Won 2nd Place at World's Largest Hardware Hackathon

- Developed **ESP32** firmware for a BLE haptic vest, translating AI perception commands into real-time motor actuation.
- Implemented a **9-field BLE JSON protocol** with packet timeouts, priority arbitration, and mode-specific control.
- Integrated **ultrasonic proximity sensing** to trigger local obstacle safety overrides independent of iPhone perception.
- Validated **4 navigation modes**: awareness, object search, GPS guidance, and manual haptic testing.

RISC-V Operating System Kernel

- Built a **RISC-V OS kernel** in **C** supporting user-mode execution, threading, traps, syscalls, and process management.
- Implemented **VirtIO-backed storage** with NGFS filesystem support, block caching, persistent I/O, and directory operations.
- Developed **Sv39 virtual memory** with 3-level page-tables, process address-space cloning, and lazy page allocation.
- Integrated ELF loading, scheduling, interrupt/exception handling, **UART I/O**, and shell support for disk program execution.

CUDA GPT-2 Inference Optimizer

- Optimized kernels for inference across attention, layernorm, softmax, GELU, residuals, and matmul, achieving **9.7x speedup**.
- Profiled workloads with nsys and ncu, analyzing kernel timing, memory traffic, cache behavior, and occupancy to find bottlenecks.

Guitar Auto-Tuner

- Designed a guitar auto-tuner using **band-pass filters** and **peak detection**.
- Built **RLC filters at 630 Hz, 660 Hz, and 690 Hz** to isolate string tones, simulated in **LTSpice**.
- Implemented **comparator circuits** and an **H-bridge motor driver** to adjust tuning pegs automatically.
- Prototyped and tested circuits on breadboards, validating tuning accuracy with oscilloscopes and audio input.

Illini Electric Vehicle Concept

- Designed **2 active clamp power converters** using MasterGaN3 ICs, converting **48 V** to **5 V/12 V** at **95% efficiency**.
- Researched components, designed PCBs in KiCad, while developing firmware on an **STM32 microcontroller**.

Skills

Programming Languages: C, C++, Python, PyTorch, CUDA, SystemVerilog, RISC-V, Java

Frameworks and Tools: Linux, I2C, SPI, CAN bus, STM32, ESP32, Raspberry Pi, FreeRTOS, ROS, SES, Keil µVision, Synopsys VCS, Verdi, DVE, LTSpice, NumPy, KiCad, Git, GitHub, GitLab, Vivado, Vim